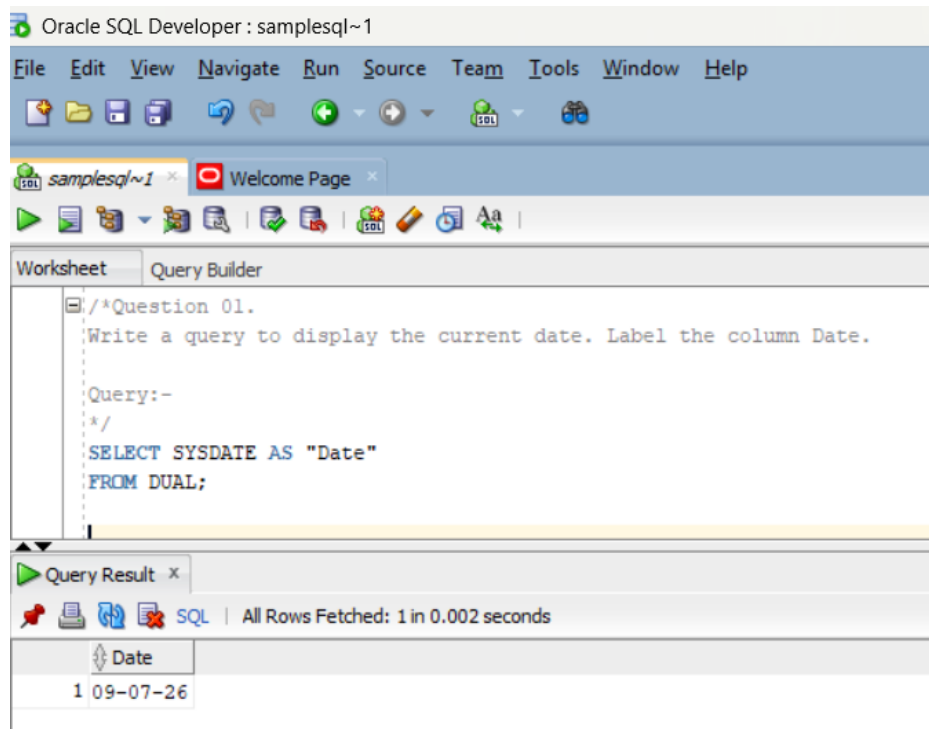


USING SINGLE-ROW FUNCTION

Practice Assessment Questions---

01.



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Worksheet Query Builder

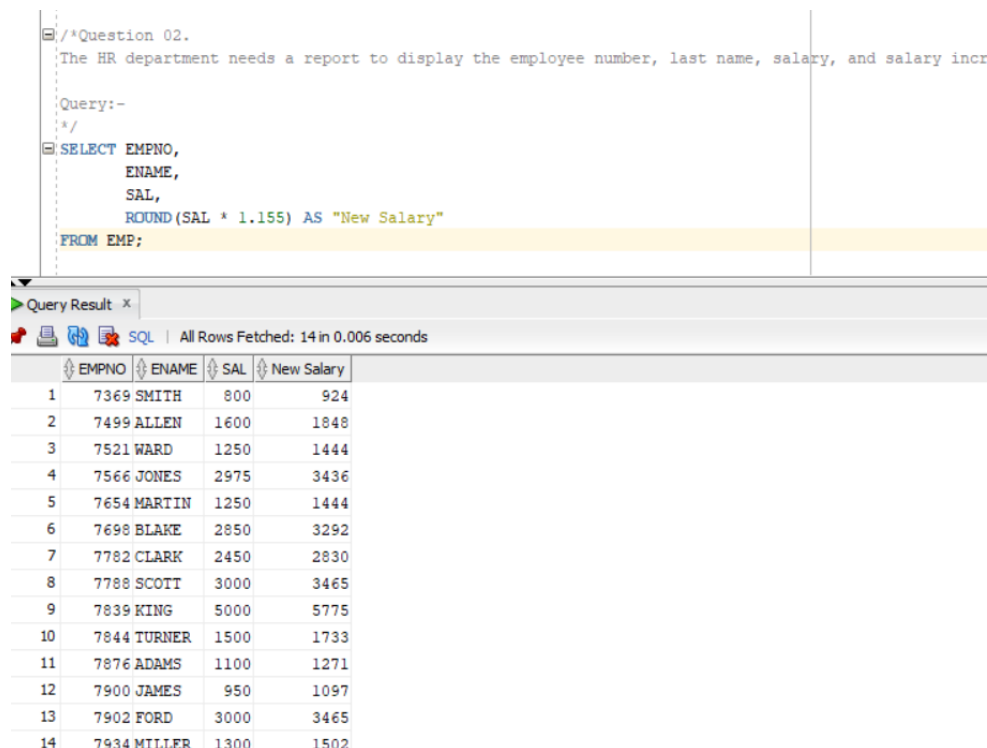
```
/*Question 01.  
Write a query to display the current date. Label the column Date.  
  
Query:-  
*/  
SELECT SYSDATE AS "Date"  
FROM DUAL;
```

Query Result x

SQL | All Rows Fetched: 1 in 0.002 seconds

	Date
1	09-07-26

02.



```
/*Question 02.  
The HR department needs a report to display the employee number, last name, salary, and salary incr  
  
Query:-  
*/  
SELECT EMPNO,  
       ENAME,  
       SAL,  
       ROUND(SAL * 1.155) AS "New Salary"  
FROM EMP;
```

Query Result x

SQL | All Rows Fetched: 14 in 0.006 seconds

	EMPNO	ENAME	SAL	New Salary
1	7369	SMITH	800	924
2	7499	ALLEN	1600	1848
3	7521	WARD	1250	1444
4	7566	JONES	2975	3436
5	7654	MARTIN	1250	1444
6	7698	BLAKE	2850	3292
7	7782	CLARK	2450	2830
8	7788	SCOTT	3000	3465
9	7839	KING	5000	5775
10	7844	TURNER	1500	1733
11	7876	ADAMS	1100	1271
12	7900	JAMES	950	1097
13	7902	FORD	3000	3465
14	7934	MILLER	1300	1502

03.

```
/*Question 03.
Modify the previous query to add a column alias that subtracts the old salary from the new salary. Label the column Increase.

Query:-
*/
SELECT EMPNO,
       ENAME,
       SAL,
       ROUND(SAL * 1.155) AS "New Salary",
       ROUND(SAL * 1.155) - SAL AS "Increase"
FROM EMP;
```

Query Result x

All Rows Fetched: 14 in 0.006 seconds

EMPNO	ENAME	SAL	New Salary	Increase
1	7369 SMITH	800	924	124
2	7499 ALLEN	1600	1848	248
3	7521 WARD	1250	1444	194
4	7566 JONES	2975	3436	461
5	7654 MARTIN	1250	1444	194
6	7698 BLAKE	2850	3292	442
7	7782 CLARK	2450	2830	380
8	7788 SCOTT	3000	3465	465
9	7839 KING	5000	5775	775
10	7844 TURNER	1500	1733	233
11	7876 ADAMS	1100	1271	171
12	7900 JAMES	950	1097	147
13	7902 FORD	3000	3465	465
14	7934 MILLER	1300	1502	202

04.

```
/*Question 04.
Write a query that displays the ename (with the first letter uppercase and all other letters lowercase) and the length of the name.

Query:-
*/
SELECT INITCAP(ENAME) AS "Employee Name",
       LENGTH(ENAME) AS "Length"
FROM EMP
WHERE ENAME LIKE 'J%'
      OR ENAME LIKE 'A%'
      OR ENAME LIKE 'M%'
ORDER BY ENAME;
```

Query Result x

All Rows Fetched: 6 in 0.022 seconds

Employee Name	Length
1 Adams	5
2 Allen	5
3 James	5
4 Jones	5
5 Martin	6
6 Miller	6

05.

```
/*Question 05.
Rewrite the query so that the user is prompted to enter a letter that starts the last name. For e

Query:-
*/
SELECT INITCAP(ENAME) AS "Employee Name",
       LENGTH(ENAME) AS "Length"
FROM EMP
WHERE ENAME LIKE 'A%'
ORDER BY ENAME;
```

Query Result x

SQL | All Rows Fetched: 2 in 0.004 seconds

	Employee Name	Length
1	Adams	5
2	Allen	5

06.

```
/*Question 06.
The HR department wants to find the length of employment for each employee. For each employee, displa

Query:-
*/
SELECT ENAME,
       ROUND(MONTHS_BETWEEN(SYSDATE, HIREDATE)) AS MONTHS_WORKED
FROM EMP
ORDER BY MONTHS_WORKED;
```

Query Result x

SQL | All Rows Fetched: 14 in 0.004 seconds

	ENAME	MONTHS_WORKED
1	ADAMS	470
2	SCOTT	471
3	MILLER	534
4	FORD	535
5	JAMES	535
6	KING	536
7	MARTIN	537
8	TURNER	538
9	CLARK	541
10	BLAKE	542
11	JONES	543
12	WARD	545
13	ALLEN	545
14	SMITH	547

07.

```
/*Question 07.
Create a report that produces the following for each employee: earns monthly but wants <3 times salary>. Label the column Dream Salaries.

Query:-
*/

SELECT ENAME || ' earns monthly but wants ' ||
       TO_CHAR(SAL * 3) AS "Dream Salaries"
FROM EMP;
```

Query Result x

SQL | All Rows Fetched: 14 in 0.005 seconds

	Dream Salaries
1	SMITH earns monthly but wants 2400
2	ALLEN earns monthly but wants 4800
3	WARD earns monthly but wants 3750
4	JONES earns monthly but wants 8925
5	MARTIN earns monthly but wants 3750
6	BLAKE earns monthly but wants 8550
7	CLARK earns monthly but wants 7350
8	SCOTT earns monthly but wants 9000
9	KING earns monthly but wants 15000
10	TURNER earns monthly but wants 4500
11	ADAMS earns monthly but wants 3300
12	JAMES earns monthly but wants 2850
13	FORD earns monthly but wants 9000
14	MILLER earns monthly but wants 3900

08.

```
/*Question 08.
Create a query to display the last name and salary for all employees. Format the salary

Query:-
*/

SELECT ENAME,
       LPAD(SAL, 15, '$') AS "SALARY"
FROM EMP;
```

Query Result x

SQL | All Rows Fetched: 14 in 0.003 seconds

	ENAME	SALARY
1	SMITH	\$\$\$\$\$\$\$\$\$\$\$\$800
2	ALLEN	\$\$\$\$\$\$\$\$\$\$\$\$1600
3	WARD	\$\$\$\$\$\$\$\$\$\$\$\$1250
4	JONES	\$\$\$\$\$\$\$\$\$\$\$\$2975
5	MARTIN	\$\$\$\$\$\$\$\$\$\$\$\$1250
6	BLAKE	\$\$\$\$\$\$\$\$\$\$\$\$2850
7	CLARK	\$\$\$\$\$\$\$\$\$\$\$\$2450
8	SCOTT	\$\$\$\$\$\$\$\$\$\$\$\$3000
9	KING	\$\$\$\$\$\$\$\$\$\$\$\$5000
10	TURNER	\$\$\$\$\$\$\$\$\$\$\$\$1500
11	ADAMS	\$\$\$\$\$\$\$\$\$\$\$\$1100
12	JAMES	\$\$\$\$\$\$\$\$\$\$\$\$950
13	FORD	\$\$\$\$\$\$\$\$\$\$\$\$3000
14	MILLER	\$\$\$\$\$\$\$\$\$\$\$\$1300

09.

```

/*Question 09.
Display each employee's last name, hire date, and salary review date, which is the first
Query:-
*/
SELECT ENAME,
       HIREDATE,
       TO_CHAR(
         NEXT_DAY(ADD_MONTHS(HIREDATE,6), 'MONDAY'),
         'Day, "the" DD "of" Month, YYYY'
       ) AS REVIEW
FROM EMP;

```

Query Result x

SQL | All Rows Fetched: 14 in 0.004 seconds

	ENAME	HIREDATE	REVIEW
1	SMITH	17-12-80	Monday , the 22 of June , 1981
2	ALLEN	20-02-81	Monday , the 24 of August , 1981
3	WARD	22-02-81	Monday , the 24 of August , 1981
4	JONES	02-04-81	Monday , the 05 of October , 1981
5	MARTIN	28-09-81	Monday , the 29 of March , 1982
6	BLAKE	01-05-81	Monday , the 02 of November , 1981
7	CLARK	09-06-81	Monday , the 14 of December , 1981
8	SCOTT	19-04-87	Monday , the 26 of October , 1987
9	KING	17-11-81	Monday , the 24 of May , 1982
10	TURNER	08-09-81	Monday , the 15 of March , 1982
11	ADAMS	23-05-87	Monday , the 30 of November , 1987
12	JAMES	03-12-81	Monday , the 07 of June , 1982
13	FORD	03-12-81	Monday , the 07 of June , 1982
14	MILLER	23-01-82	Monday , the 26 of July , 1982

10.

```

/*Question 10.
Display the last name, hire date, and day of the week on which the employee started. Label the column DAY. Order the results by the day of the week, starting with Monday.
Query:-
*/
SELECT ENAME,
       HIREDATE,
       TO_CHAR(HIREDATE, 'DAY') AS DAY
FROM EMP
ORDER BY TO_CHAR(HIREDATE, 'D');

```

Query Result x

SQL | All Rows Fetched: 14 in 0.003 seconds

	ENAME	HIREDATE	DAY
1	SCOTT	19-04-87	SUNDAY
2	WARD	22-02-81	SUNDAY
3	MARTIN	28-09-81	MONDAY
4	CLARK	09-06-81	TUESDAY
5	TURNER	08-09-81	TUESDAY
6	KING	17-11-81	TUESDAY
7	SMITH	17-12-80	WEDNESDAY
8	JAMES	03-12-81	THURSDAY
9	FORD	03-12-81	THURSDAY
10	JONES	02-04-81	THURSDAY
11	BLAKE	01-05-81	FRIDAY
12	ALLEN	20-02-81	FRIDAY
13	MILLER	23-01-82	SATURDAY
14	ADAMS	23-05-87	SATURDAY

11.

```

/*Question 11.
Create a query that displays the employees' last names and commission amounts. If an employee does not earn commission, show "No Commission." Label the column COMM.

Query:-
*/
SELECT ENAME,
       NVL(TO_CHAR(COMM), 'No Commission') AS COMM
FROM EMP;

```

Query Result x

SQL | All Rows Fetched: 14 in 0.008 seconds

	ENAME	COMM
1	SMITH	No Commission
2	ALLEN	300
3	WARD	500
4	JONES	No Commission
5	MARTIN	1400
6	BLAKE	No Commission
7	CLARK	No Commission
8	SCOTT	No Commission
9	KING	No Commission
10	TURNER	0
11	ADAMS	No Commission
12	JAMES	No Commission
13	FORD	No Commission
14	MILLER	No Commission

12.

```

/*Question 12.
Create a query that displays the first eight characters of the employees' last name

Query:-
*/
SELECT SUBSTR(ENAME,1,8) || ' ' ||
       RPAD(' ', ROUND(SAL/1000), ' ')
       AS EMPLOYEES_AND_THEIR_SALARIES
FROM EMP
ORDER BY SAL DESC;

```

Query Result x

SQL | All Rows Fetched: 14 in 0.004 seconds

	EMPLOYEES_AND_THEIR_SALARIES
1	KING *****
2	FORD ***
3	SCOTT ***
4	JONES ***
5	BLAKE ***
6	CLARK **
7	ALLEN **
8	TURNER **
9	MILLER *
10	WARD *
11	MARTIN *
12	ADAMS *
13	JAMES *
14	SMITH *

13.

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Worksheet Query Builder

/*Question 13.
Using the DECODE function, write a query that displays the grade of all employees based on the value of the column JOB_ID

Query:-
*/
SELECT ENAME,
 JOB,
 DECODE (JOB,
 'PRESIDENT','A',
 'MANAGER','B',
 'SALESMAN','C',
 'CLERK','D',
 'Not Assigned') AS GRADE
FROM EMP;

Query Result x

SQL | All Rows Fetched: 14 in 0.004 seconds

	ENAME	JOB	GRADE
1	SMITH	CLERK	D
2	ALLEN	SALESMAN	C
3	WARD	SALESMAN	C
4	JONES	MANAGER	B
5	MARTIN	SALESMAN	C
6	BLAKE	MANAGER	B
7	CLARK	MANAGER	B
8	SCOTT	ANALYST	Not Assigned
9	KING	PRESIDENT	A
10	TURNER	SALESMAN	C
11	ADAMS	CLERK	D
12	JAMES	CLERK	D
13	FORD	ANALYST	Not Assigned
14	MILLER	CLERK	D

14.

The screenshot shows a SQL query editor window with a toolbar at the top. The 'Worksheet' tab is active, and the 'Query Builder' sub-tab is selected. The query text is as follows:

```
/*Question 14.  
Rewrite the statement in the preceding exercise using the CASE syntax.  
  
Query:-  
*/  
SELECT ENAME,  
       JOB,  
       CASE  
         WHEN JOB='PRESIDENT' THEN 'A'  
         WHEN JOB='MANAGER' THEN 'B'  
         WHEN JOB='SALESMAN' THEN 'C'  
         WHEN JOB='CLERK' THEN 'D'  
         ELSE 'Not Assigned'  
       END AS GRADE  
FROM EMP;
```

Below the query editor, the 'Query Result' window is open, showing the results of the query. The status bar indicates 'All Rows Fetched: 14 in 0.006 seconds'. The results are displayed in a table with three columns: ENAME, JOB, and GRADE.

	ENAME	JOB	GRADE
1	SMITH	CLERK	D
2	ALLEN	SALESMAN	C
3	WARD	SALESMAN	C
4	JONES	MANAGER	B
5	MARTIN	SALESMAN	C
6	BLAKE	MANAGER	B
7	CLARK	MANAGER	B
8	SCOTT	ANALYST	Not Assigned
9	KING	PRESIDENT	A
10	TURNER	SALESMAN	C
11	ADAMS	CLERK	D
12	JAMES	CLERK	D
13	FORD	ANALYST	Not Assigned
14	MILLER	CLERK	D